

Analysis PhD Exam, May 2012

Be sure to carefully present all work in a neat and logical fashion. Do not leave any gaps. State clearly theorems used in your proofs.

1. Let μ be a real-valued signed measure on a σ -algebra Σ . Does there exist a set in Σ on which μ attains its minimum value? Prove or disprove.
2. Let X be a Banach space. Show that there exists a compact T_2 space S such that X is isometrically isomorphic to a closed linear subspace of $C(S)$. [Hint: Examine the unit ball of the dual space X^* with the appropriate topology and the action of X on X^* .]
3. Let $\{f_n\}$ be a sequence of integrable functions on S such that $\sum \int |f_n| d\mu < \infty$. Give all details concerning the convergence of $\sum f_n(s)$, $s \in S$.
4. State and prove the Fubini theorem.
5. Let π be the natural map from the Banach space X into its second dual X^{**} . Show $\pi(X)$ is closed in the norm topology of X^{**} .
6. Show that the continuous functions on $[0, 1]$ are dense in $L^1[0, 1]$ (Lebesgue measure). [Hint: First show this is true for the indicator function of a Lebesgue measurable subset of $[0, 1]$.]